

2.3.2

~2068

$$\int \frac{dx}{x(x^5+5x^2)} = \left[x=t^{10} \right] = \int \frac{10t^9 dt}{t^{10}(t^5+t^4)} = 10 \int \frac{t^9 dt}{t^5(t^5+t^4)} =$$

$$= 10 \int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t^3} + \frac{D}{t^4} + \frac{E}{t^5} + \frac{F}{t+1} \right) dt \quad \textcircled{2}$$

$$A t^4(t+1) + B t^3(t+1) + C t^2(t+1) + D t(t+1) + E(t+1) + F t^5 = 1$$

$$\begin{cases} A t^5 + F t^5 = 0 t^2 \\ A t^4 + B t^4 = 0 \\ A t^3 + C t^3 = 0 \\ C t^2 + D t^2 = 0 \\ D t + E t = 0 \\ E = 1 \end{cases} \Rightarrow \begin{cases} E = 1 \\ D = -1 \\ C = 1 \\ B = -1 \\ A = 1 \\ F = -1 \end{cases}$$

$$\textcircled{2} 10 \int \left(\frac{1}{t} + \frac{-1}{t^2} + \frac{1}{t^3} + \frac{-1}{t^4} + \frac{1}{t^5} + \frac{-1}{t+1} \right) dt =$$

$$= 10 \left(\ln|t| + \frac{1}{t} - \frac{1}{2t^2} + \frac{1}{3t^3} - \frac{1}{4t^4} - \ln|t+1| \right) =$$

$$= 10 \ln|\sqrt[10]{x}| + \frac{10}{\sqrt[10]{x}} - \frac{5}{5\sqrt{x}} + \frac{10}{3\sqrt[3]{x^3}} - \frac{5}{2\sqrt[5]{x^5}} - 10 \ln|\sqrt[10]{x}+1| + C$$

~2069

$$\int \frac{dx}{x^2+5x^2+2\sqrt{x}} = \left[x=t^{12} \right] = \int \frac{12t^{11} dt}{t^{12}+t^9+2t^3} = 12 \int \frac{t^8 dt}{t^3+t^2+2} =$$

$$= 12 \int \left(\frac{(t^3+t+2)(t^5-t^3-t^2+t+3)}{t^3+t+2} + \frac{t^2-8t-6}{t^3+t+2} \right) dt =$$

$$= 12 \int (t^2-t^3-t^2+3) dt + 12 \int \frac{(t+1)(t-6)}{(t+1)(t^2-t+1)} dt =$$

$$= 12 \cdot \frac{t^3}{3} - 12 \cdot \frac{t^4}{4} - 12 \cdot \frac{t^3}{3} + 12 \cdot 3t + 12 \int \frac{t-6}{t^2-t+1} dt =$$

$$\begin{aligned}
 &= 2t^6 - 3t^4 - 6t^3 + 6t^2 + 36t + \int \frac{t-6}{t^2-t+1} dt = 2t^6 - 3t^4 - 6t^3 + 6t^2 + 36t + \int \frac{1}{2} \frac{d(t^2-t+1)}{t^2-t+1} - \frac{11}{2} \frac{dt}{t^2-t+1} \\
 &= 2t^6 - 3t^4 - 6t^3 + 6t^2 + 36t + \frac{1}{2} \ln|t^2-t+1| - \frac{11}{4} \int \frac{dx}{t^2-2\frac{1}{2}t+\frac{5}{4}} = 2\sqrt{x} - 3x^{\frac{1}{2}} - 6x^{\frac{1}{4}} + 6x^{\frac{1}{4}} + 36x^{\frac{1}{2}} + \frac{1}{2} \ln|x^2-5x+1| - \\
 &= \frac{11}{4} \int \frac{dx}{(t-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = A - \frac{11}{4} \arctan \left| \frac{t-\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right| \cdot \frac{2}{\sqrt{3}} + C = \\
 &= A - \frac{11}{4} \arctan \left| \frac{\sqrt{x}-\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right| \cdot \frac{2}{\sqrt{3}} + C =
 \end{aligned}$$

✓ 2071

$$\int \frac{1-x}{1+x} \frac{dx}{x}$$

Заменим $\frac{ax+b}{cx+d} = t = \frac{-x+1}{x+1}$

~~Заменим~~ $\frac{-x+1}{x+1} = t^2$ ~~или~~ $x = \frac{1-t^2}{t^2+1}$

$$dx = \frac{2(-1-t)dt}{(t^2+1)^2} = \frac{2}{(t^2+1)^2} dt$$

$$\int \frac{2}{(t^2+1)^2} \cdot \frac{t^2+1}{(-t^2+1)} dt = 2 \int \frac{dt}{(t^2+1)(t^2-1)} = 2 \int \left(\frac{A}{t^2+1} + \frac{B}{t^2-1} \right) dt \quad \textcircled{2}$$

$$At^3 - At + Ct^2 - C + Bt^3 + Bt + Dt^2 + D = t^2$$

$$\begin{cases} A+B=0 \\ C+D=1 \\ -A+D=0 \\ -C+D=0 \end{cases} \Rightarrow \begin{cases} A=0 \\ B=0 \\ C=1/2 \\ D=1/2 \end{cases}$$

$$\begin{aligned}
 \textcircled{2} \quad 4 \int \left(\frac{1/2}{t^2+1} + \frac{1/2}{t^2-1} \right) dt &= 2 \int \frac{dt}{t^2+1} + 2 \int \frac{dt}{t^2-1} = 2 \arctan t + \\
 &+ \ln \left| \frac{t-1}{t+1} \right| = 2 \arctan \sqrt{\frac{1-x}{x+1}} + 2 \ln \left| \frac{\sqrt{\frac{1-x}{x+1}} - 1}{\sqrt{\frac{1-x}{x+1}} + 1} \right| + C
 \end{aligned}$$

2072

③

$$\int \sqrt{\frac{1-x}{1+x}} dx = \left[\begin{array}{l} t = \sqrt{x} \\ 1-x^2 = x \\ dx = 2t dt \end{array} \right] = \int \sqrt{\frac{1-t}{1+t}} \cdot 2t dt \quad (1)$$

Samana $\sqrt{\frac{1-t}{1+t}} = z$ ~~22~~ $z^2 = \frac{1-t}{1+t}$

$$z^2(1+t) = -t+1$$

$$z^2 t + z^2 + t - 1 = 0$$

$$t(z^2 + 1) = -z^2 + 1$$

$$t = \frac{-z^2 + 1}{z^2 + 1}$$

$$dt = \frac{-2z(z^2+1) - 2z(1-z^2)}{(1+z^2)^2}$$

$$dt = \frac{-4z}{(1+z^2)^2} dz$$

$$\textcircled{2} 2 \int z \cdot \frac{-z^2+1}{z^2+1} \cdot \frac{-4z dz}{(1+z^2)^2} = -8 \int \frac{z^2(1-z^2) dz}{(1+z^2)^3} =$$

$$= -8 \int \left(\frac{Az+B}{1+z^2} + \frac{Cz+D}{(1+z^2)^2} + \frac{Ez+F}{(1+z^2)^3} \right) dz \quad (2)$$

$$z^2 - z^4 = (Az+B)(1+z^2) + (Cz+D)(1+z^2)^2 + (Ez+F)(1+z^2)^3$$

$$Az^5 + 2Az^3 + Az + Bz^4 + 2Bz^2 + B + Cz^3 + Cz + Dz^2 + D + Ez^2 + Ez + Fz + F$$

$$\begin{cases} A=0 \\ B=1 \\ C=0 \\ D=3 \\ E=0 \\ F=-2 \end{cases}$$

$$\textcircled{2} -8 \int \left(\frac{-1}{1+z^2} + \frac{3}{(1+z^2)^2} + \frac{-2}{(1+z^2)^3} \right) dz =$$

$$= -8 \int \frac{1}{1+z^2} dz - 24 \int \frac{dz}{(1+z^2)^2} + 16 \int \frac{dz}{(1+z^2)^3} = 8 \arctg z -$$

$$-24 \cdot \frac{1}{2} \arctg z + 16 \int \frac{dz}{(1+z^2)^3} \quad (3)$$

$$A = \int \frac{du \cdot \cos^6 u}{\cos^2 u} = \int \cos^4 u du = \int (1 - \sin^2 u)^2 du \quad (4)$$

$$z = \tan u \quad \frac{dz}{du} = \frac{1}{\cos^2 u}$$

$$\begin{aligned}
 \textcircled{3} \int \left(\frac{\cos 2u + 1}{2} \right)^2 du &= \frac{1}{4} \int (\cos^2 2u + 2\cos 2u + 1) du = \\
 &= \frac{1}{4} \int \frac{\cos 4u + 1}{2} + \frac{1}{2} \int \cos 2u + \frac{1}{4} \int du = \\
 &= \frac{1}{8} \int \cos 4u du + \frac{u}{8} + \frac{1}{4} \sin 2u + \frac{1}{4} u = \\
 &= \frac{3}{8} u + \frac{1}{32} \sin 4u + \frac{1}{4} \sin 2u = \frac{3}{8} \operatorname{arctg} z + \frac{1}{32} \sin 4 \operatorname{arctg} z \\
 &+ \frac{1}{4} \sin 2 \operatorname{arctg} z = \operatorname{arctg} z -
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \operatorname{arctg} z &= \frac{z}{1+z^2} + \dots \operatorname{arctg} z \text{ for } z = \frac{z}{1+z^2} \cdot \left(\frac{1}{1+z^2} \right)^{-1} = \\
 &= \operatorname{arctg} z - \left(\frac{z}{1+z^2} \right) \left(-1 + 2 \frac{1-z^2}{1+z^2} \right) + C \\
 &\quad 2102
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{dx}{\sin^3 x} &= - \int \frac{d \operatorname{ctg} x}{\sin x} = - \frac{\operatorname{ctg} x}{\sin x} + \int \operatorname{ctg} x d \left(\frac{1}{\sin x} \right) = \\
 &= - \frac{\cos x}{\sin^2 x} - \int \frac{\cos x}{\sin x} \cdot \frac{\cos x}{\sin^2 x} dx = - \left(\frac{\cos x}{\sin^2 x} + \int \frac{1 - \sin^2 x}{\sin^3 x} dx \right) = \\
 &= - \left(\frac{\cos x}{\sin^2 x} + \int \frac{dx}{\sin^3 x} - \int \frac{dx}{\sin x} \right) = - \frac{\cos x}{\sin^2 x} + \int \frac{dx}{\sin x} - \int \frac{dx}{\sin^3 x} \textcircled{2} \\
 \int \frac{dx}{\sin x} &= \int \frac{1}{\sin x} = \frac{2t}{t^2+1} \quad dx = \frac{2dt}{1+t^2} \quad \int = \int \frac{2dt}{1+t^2} \cdot \frac{1+t^2}{2t} = \int \frac{dt}{t} = \\
 &= \ln|t| = \ln \left| \operatorname{tg} \frac{x}{2} \right|
 \end{aligned}$$

$$\textcircled{2} - \frac{\cos x}{\sin^2 x} + \ln \left| \operatorname{tg} \frac{x}{2} \right| - \underbrace{\int \frac{dx}{\sin^3 x}}_{-A} \Rightarrow A = - \frac{\cos x}{\sin^2 x} + \ln \left| \operatorname{tg} \frac{x}{2} \right| - A$$

$$2A = - \frac{\cos x}{\sin^2 x} + \ln \left| \operatorname{tg} \frac{x}{2} \right| \Rightarrow A = - \frac{\cos x}{2 \sin^2 x} + \frac{1}{2} \ln \left| \operatorname{tg} \frac{x}{2} \right| + C$$

2110

$$\begin{aligned}
 \int \frac{dx}{5-3\cos x} &= \int \frac{1}{5-3\cos x} = \int \frac{1}{5-3 \cdot \frac{1-t^2}{1+t^2}} \quad dx = \frac{2dt}{1+t^2} \quad \int = \int \frac{2dt}{1+t^2} \cdot \frac{1}{5 \cdot \frac{2t^2}{1+t^2} + 2} = \\
 &= \int \frac{dt}{1+t^2} \cdot \frac{1+t^2}{3t^2+1+t^2} = \int \frac{dt}{4t^2+1} = \frac{1}{2} \operatorname{arctg} 2t = \frac{1}{2} \operatorname{arctg} 2 \operatorname{tg} \frac{x}{2} + C
 \end{aligned}$$

2114

⑤

$$\int \frac{dx}{4 + 6 \log x + 4 \log^2 x} = \left[\begin{array}{l} t = \log x \quad x = \text{antilog } t \\ 4 \log x = \frac{1}{t} \quad dx = \frac{dt}{1+t^2} \end{array} \right] = \int \frac{dt}{1+t^2} \cdot \frac{t}{(t+2)^2} =$$

$$= \int \left(\frac{A}{t+2} + \frac{B}{(t+2)^2} + \frac{Ct+D}{t^2+1} \right) dt \quad \textcircled{2}$$

$$\frac{A}{t+2} + \frac{B}{(t+2)^2} + \frac{Ct+D}{t^2+1} = \frac{t}{(t+2)^2(1+t^2)} \quad | \quad (t+2)^2(1+t^2)$$

$$A(t+2)(t^2+1) + B(t^2+1) + (Ct+D)(t+2)^2 = t$$

$$t^3(C+A) + t^2(4C+D+2A+B) + t(4C+4D+A) + (4D+2A+B) = t$$

$$\left\{ \begin{array}{l} A = -C \\ B = -2A - 4C - D \\ 3C = 1 - 4D \\ B = -2A - 4D \end{array} \right. \quad \Rightarrow \quad \left\{ \begin{array}{l} A = -C \\ B = -2C - D \\ 3C = 1 - 4D \\ -2(1-4D) - D = +2 \cdot \frac{1}{3}(1-4D) - 4D \end{array} \right.$$

$$\left\{ \begin{array}{l} A = -C = -\frac{3}{25} \\ B = -\frac{6}{25} - \frac{4}{25} - \frac{10}{25} = -\frac{2}{5} \\ C = \frac{1}{3} \left(1 - \frac{16}{25} \right) = \frac{3}{25} \\ D = \frac{4}{25} \end{array} \right.$$

$$\textcircled{2} \int \left(-\frac{3}{25} \cdot \frac{1}{t+2} - \frac{2}{5(t+2)^2} + \frac{3t+4}{25(t^2+1)} \right) dt = -\frac{3}{25} \int \frac{dt}{t+2}$$

$$- \frac{2}{5} \int \frac{dt}{(t+2)^2} + \frac{1}{25} \int \frac{3t+4}{t^2+1} dt = -\frac{3}{25} \cdot \ln|t+2| - \frac{2}{5} \cdot \frac{(t+2)^{-2+1}}{-2+1}$$

$$+ \frac{1}{25} \int \frac{3t dt}{t^2+1} + \frac{4}{25} \int \frac{dt}{t^2+1} = -\frac{3}{25} \ln|t+2| + \frac{2}{5(t+2)} +$$

$$+ \frac{3}{25} \cdot \frac{1}{2} \int \frac{dt^2}{t^2+1} + \frac{4}{25} \cdot \arctan t = -\frac{3}{25} \ln|\log x + 2| + \frac{2}{5(\log x + 2)} +$$

$$+ \frac{4}{25} \cdot x + \frac{3}{50} \ln|\log^2 x + 1| = -\frac{3}{25} \ln|\log x + 2| + \frac{2}{5(\log x + 2)} + \frac{4}{25} x + \frac{3}{50} \ln \frac{1}{\cos^2 x}$$

x^c

2103 ⑥

$$\int \frac{\cos^4 x + \sin^4 x}{\cos^2 x - \sin^2 x} dx = \int \frac{(\cos^2 x + \sin^2 x)^2 - 2\cos^2 x \sin^2 x}{\cos 2x} dx = \int \frac{1 - \frac{1}{2} \sin^2 2x}{\cos 2x} dx$$

$$dx = \frac{1}{4} \int \frac{2 - \sin^2 2x}{\cos 2x} d(2x) = \frac{1}{4} \int \frac{1 + \cos^2 a}{\cos a} da = \frac{1}{4} \int \frac{da}{\cos a} +$$

$$+ \frac{1}{4} \int \cos a da = \frac{1}{4} \int \frac{da}{\cos a} + \frac{1}{4} \int \cos 2x d2x = \frac{1}{4} \sin 2x +$$

$$+ \frac{1}{4} \int \frac{da}{\cos a} = \left[t = \tan \frac{a}{2} = \tan x \quad \cos a = \frac{1-t^2}{1+t^2} \right] = \frac{1}{4} \sin 2x + \frac{1}{4} \int \frac{2dt}{1+t^2}$$

$$\cdot \frac{1+t^2}{1-t^2} = \frac{1}{4} \sin 2x + \frac{1}{2} \int \frac{dt}{1-t^2} = \frac{1}{4} \sin 2x + \frac{1}{2} \ln \left| \frac{1+t}{1-t} \right| = \frac{1}{4} \sin 2x + \frac{1}{4} \cdot$$

$$\ln \left| \frac{1+\tan x}{1-\tan x} \right| + C$$

2090

$$\int \sin^3 x \cos^2 x dx = \int \cos^2 x (1 - \cos^2 x) d(\sin x) = \int (\cos^2 x - \cos^4 x) d\cos x = \int \cos^2 x d\cos x - \int \cos^4 x d\cos x = \frac{1}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C$$

2091

$$\int \frac{\sin^3 x}{\cos^4 x} dx = \int \frac{1 - \cos^2 x}{\cos^4 x} d(\sin x) = \int \frac{\cos^2 x - 1}{\cos^4 x} d\cos x = \int \frac{d\cos x}{\cos^2 x} - \int \frac{d\cos x}{\cos^4 x} = \frac{\cos^{2+1} x}{-1} + \frac{(\cos x)^{-4+1}}{-3} = \frac{3}{\cos^3 x} - \frac{1}{\cos x} + C$$

2092

$$\int \cot^4 x dx = \int \frac{\cos^4 x}{\sin^4 x} dx = \int \frac{(1 - \sin^2 x)^2}{\sin^4 x} dx = \int \frac{1 - 2\sin^2 x + \sin^4 x}{\sin^4 x} dx$$

$$dx = \int \frac{dx}{\sin^4 x} - 2 \int \frac{dx}{\sin^2 x} + \int dx = x + 2 \cot x - \frac{1}{3} \cot^3 x + C$$

2152

$$\int \frac{dx}{x\sqrt{x^2+4x-4}} = \left[\frac{1}{x} = t \right] = \int t \cdot \frac{-dt}{t^2} \cdot \frac{1}{\sqrt{\frac{1}{t^2} + \frac{4}{t} - 4}} =$$

$$= - \int \frac{1}{t} \cdot \frac{t dt}{\sqrt{-4t^2+4t+1}} = - \int \frac{dt}{(2t-1)^2-2} = - \frac{1}{2} \int \frac{d(2t-1)}{\sqrt{(2t-1)^2-2}} =$$

$$= - \frac{1}{2} \arcsin \frac{2t-1}{\sqrt{2}} = - \frac{1}{2} \arcsin \left(\frac{\sqrt{2}}{x} - \frac{1}{\sqrt{2}} \right) + C$$

2160

$$\begin{aligned} \int \sqrt{1-4x-x^2} dx &= \int \sqrt{(x+2)^2-5} d(x+2) = \int \sqrt{5-A^2} dA = \\ &= \sqrt{5-A^2} \cdot A - \int A d(\sqrt{5-A^2}) = A\sqrt{5-A^2} - \int A \cdot \frac{-2A}{2\sqrt{5-A^2}} dA = \\ &= A\sqrt{5-A^2} + \int \frac{A^2 dA}{\sqrt{5-A^2}} = A\sqrt{5-A^2} - \int \sqrt{5-A^2} dA + 5 \int \frac{dA}{\sqrt{5-A^2}} = \\ &= A\sqrt{5-A^2} + 5 \arcsin \frac{A}{\sqrt{5}} - \int \sqrt{5-A^2} dA \end{aligned}$$

$$2 \int \sqrt{5-A^2} dA = A\sqrt{5-A^2} + 5 \arcsin \frac{A}{\sqrt{5}}$$

$$\int \sqrt{1-4x-x^2} dx = \frac{1}{2} \cdot (x+2) \sqrt{1-4x-x^2} + 5 \arcsin \frac{x+2}{\sqrt{5}} + C$$